EE618

Assignment-2, Part-I

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1 Hand Calculations

1.1 Specifications

- $|A_v| >= 26.02$ dB $= 20$ V/V
- $V_{out} = 0.5$V
- $f_{3\text{dB}} > 100$ MHz
- $f_L \approx 1$ MHz, $f_H \geq 100$MHz
- $V_{ov} \geq 0.15V$

1.2 Deriving all the Component Parameters

As we want, $V_{ov} \geq 0.15V$ for NMOS and PMOS, I chose the starting point as $V_{ov} = 0.15V$ for NMOS.

No DC current flows through C_{large} or R_{large}, so $V_{G1} = V_{BIASN}$, and with $V_S = V_{SB} = 0$ the threshold is $V_{THn} = 0.33$ V. Hence

$$V_{BIASN} = V_{GS1} = V_{THn} + V_{ov,n} = 0.33 \text{ V} + 0.15 \text{ V} = 0.48 \text{ V}$$

Also,

$$I_D = \tfrac{1}{2}\mu C_{ox}\frac{W}{L}V_{ov}^2\,(1+\lambda V_{DS}), \qquad g_m = \frac{2I_D}{V_{ov}\,(1+\lambda V_{DS})} \approx \frac{2I_D}{V_{ov}},$$

$$g_{ds} = \frac{\lambda I_D}{1+\lambda V_{DS}}.$$

The PMOS gate is at AC ground, so it contributes only g_{ds0}. The gain from the gate of M1 to the output is

$$A_{v,g} = -g_m\,(r_{o1} \parallel r_{o0}) = -\frac{g_m}{g_{ds1}+g_{ds0}}.$$

Neglecting the $(1+\lambda V_{DS})$ factors, I_D cancels:

$$|A_{v,g}| \approx \frac{2}{V_{ov,n}\,(\lambda_n+\lambda_p)}$$

Requiring $|A_{v,g}| \geq 20$ with $V_{ov,n} = 0.15$ V:

$$\lambda_n + \lambda_p \leq \frac{2}{0.15 \times 20} = 0.667 \text{ V}^{-1}.$$

Since λ depends on L, the gain requirement fixes the channel lengths. To leave margin for the input divider loss and the neglected factors, I took $\lambda_n + \lambda_p = 0.5V^{-1}$. Choose L_n=340nm.

$$\lambda_n = 0.12 + \frac{0.065}{0.34} = 0.311 \text{ V}^{-1}, \qquad \lambda_p = 0.5 - 0.311 = 0.189 \text{ V}^{-1}$$

$$0.06 + \frac{0.04}{L_p} = 0.189 \;\Rightarrow\; L_p = 0.31\ \mu\text{m}$$

For $V_{out} = 0.5$V, we have $V_{DS1} = V_{SD0} = 0.5$ V, so

$$g_{ds1} = \frac{\lambda_n I_D}{1 + 0.5\lambda_n} = 0.2693\, I_D, \qquad g_{ds0} = \frac{\lambda_p I_D}{1 + 0.5\lambda_p} = 0.1727\, I_D,$$
$$g_{out} = 0.4420\, I_D, \qquad g_m = \frac{2I_D}{V_{ov,n}} = 13.33\, I_D,$$

$$|A_{v,g}| = \frac{g_m}{g_{out}} = \frac{13.33}{0.4420} = 30.2 \text{ V/V } (29.6 \text{ dB}),$$

which satisfies the constraint of gain > 26.02dB.

To get the widths of NMOS and PMOS, further, assume $V_{ov,p}$ also as 0.15V. Thus, saturation equations give:

$$W_n = \frac{2I_D L_n}{\mu_n C_{ox} V_{ov,n}^2 (1 + 0.5\lambda_n)} = \frac{2I_D(0.34)}{(480\ \mu)(0.15)^2(1.156)} = 5.45 \times 10^4\, I_D\ \mu\text{m},$$
$$W_p = \frac{2I_D L_p}{\mu_p C_{ox} V_{ov,p}^2 (1 + 0.5\lambda_p)} = \frac{2I_D(0.31)}{(150\ \mu) V_{ov,p}^2 (1.0945)} = 1.68 \times 10^5\, I_D\ \mu\text{m}.$$

Now, note that, since $f_H \geq 100$MHz, let's consider the minimum, that is $f_H = 100$MHz. So, with $W_n + W_p = 2.223 \times 10^5\, I_D\ \mu$m, the output capacitance is $C_{L,tot} = C_L + 0.85\,\text{fF}/\mu\text{m}\,(W_n + W_p) = 100$ fF $+ 1.89 \times 10^{-10}\, I_D$, so

$$f_H = \frac{0.442\, I_D}{2\pi\,(10^{-13} + 1.89 \times 10^{-10}\, I_D)} = 100 \text{ MHz}$$

Thus,

$$I_D = \frac{2\pi(10^8)(10^{-13})}{0.442 - 2\pi(10^8)(1.89 \times 10^{-10})} = \frac{6.283 \times 10^{-5}}{0.3233} = 194\ \mu\text{A}$$

$$W_n = 5.45 \times 10^4\, I_D = 5.45 \times 10^4 (194.4 \times 10^{-6}) = 10.6\ \mu\text{m},$$
$$W_p = 1.68 \times 10^5\, I_D = 1.68 \times 10^5 (194.4 \times 10^{-6}) = 32.6\ \mu\text{m}$$

At DC no current flows through C_{large} or into the gate, so $V_{G1} = V_{BIASN}$. For AC, R_{large} returns the gate to V_{BIASN} (an AC ground) and the gate sees C_{in} to ground:

$$\frac{V_g}{V_{in}} = \frac{sR_{large}C_{large}}{1 + sR_{large}\,(C_{large} + C_{in})}$$

This gives the mid-band divider and the lower cutoff

$$\frac{V_g}{V_{in}}\Big|_{mid} = \frac{C_{large}}{C_{large} + C_{in}}, \qquad f_L = \frac{1}{2\pi R_{large}\,(C_{large} + C_{in})}.$$

With $W_n = 10.6\ \mu$m, $L_n = 0.34\ \mu$m and $|A_{v,g}| = 30.2$:

$$C_{gs1} = \tfrac{2}{3} W_n L_n C_{ox} + W_n C_{ov} = \tfrac{2}{3}(10.6)(0.34)(18) + 10.6(0.35) = 43.3 + 3.7 = 47.0 \text{ fF},$$
$$C_{gd1} = W_n C_{ov} = 3.7 \text{ fF},$$
$$C_{in} = C_{gs1} + C_{gd1}\,(1 + |A_{v,g}|) = 47.0 + 3.7(31.2) \approx 163 \text{ fF}.$$

The overall gain is $|A_v| = |A_{v,g}| \dfrac{C_{large}}{C_{large} + C_{in}} \geq 20$ V/V (26 dB), so

$$\frac{C_{large}}{C_{large} + C_{in}} \geq \frac{20}{30.2} = 0.662 \;\Rightarrow\; C_{large} \geq 1.96\, C_{in} = 0.32 \text{ pF}.$$

Only the product $R_{large}(C_{large}+C_{in})$ is fixed by f_L, so any C_{large} above this bound works. I chose $C_{large} = 4$ pF.

$$R_{large} = \frac{1}{2\pi f_L\,(C_{large} + C_{in})} = \frac{1}{2\pi(1\text{MHz})(4.163\text{pF})} = 38.2 \text{ k}\Omega$$

1.3 Final Hand-Calculated Component Values

- $L_n = 340$ nm, $L_p = 310$ nm
- $W_n = 10.6$ μm, $W_p = 32.6$ μm
- $V_{BIASN} = 0.48$ V, $V_{BIASP} = 0.50$ V
- $C_{large} = 4$ pF, $R_{large} = 38.2$ kΩ

1.4 Component Values used for Simulation

- $L_n = 340$ nm, $L_p = 380$ nm
- $W_n = 10.59$ μm, $W_p = 19.13$ μm
- $V_{BIASN} = 0.48$ V, $V_{BIASP} = 0.447$ V
- $C_{large} = 4$ pF, $R_{large} = 40$ kΩ

1.5 Explanation of the Deviation Between Hand-Calculated Values and Values used for Simulation

While most of the parameters match, some deviations can be observed in the hand-calculated values as compared to the ones used in simulation. The hand calculations use the square-law model with the constant parameters of the table, so they give a starting point. The simulated values differ for the following reasons.

- **L_p (310 nm $\rightarrow$ 380 nm):** The hand calculation only gave the bound $L_p \geq 0.31$ μm from $\lambda_n + \lambda_p \leq 0.5$. In simulation a longer $L_p = 380$ nm was used, which lowers λ_p from 0.189 to 0.165 V^{-1}. This raises r_{o0} and the gain, and gives margin against the CLM spread of the real model.
- **V_{BIASP} and W_p (0.50 V, 32.6 μm $\rightarrow$ 0.447 V, 19.13 μm):** Both gates are fixed, so the DC output voltage is set only by the balance $I_{D1} = I_{D0}$ and is very sensitive to small errors in V_{th}, μC_{ox} and λ. The real transistor model deviates from the square law (velocity saturation, mobility degradation, V_{th} roll-off at short L), so V_{BIASP} had to be adjusted in simulation to bring the output back to its target. The lower V_{BIASP} raises the PMOS overdrive to $V_{ov,p} = 1 - 0.447 - 0.35 \approx 0.2$ V, so the same current needs a much smaller width ($W_p/L_p = 50.3$ instead of 105). The smaller

W_p also reduces the output capacitance and gives extra f_H margin, since the hand design sat exactly at 100 MHz.

- **Other model effects:** The hand calculation treats μC_{ox}, λ and the capacitances as constants. The simulator uses bias-dependent C_{db}, C_{gs} and λ and includes layout parasitics, so gain and f_H will differ by a few percent from the predicted values. This is also why the simulated design needed to be re-tuned (mainly V_{BIASP}) to meet the specs.

1.6 Bode Plot from Hand-Calculated Values

I used the following equation for the transfer function of the circuit:

$$H(s) = \underbrace{\frac{sR_0C_0}{1+sR_0C_0}}_{\text{high-pass (input node)}} \cdot \underbrace{\frac{A_v}{1+sR_{out}C_L}}_{\text{low-pass (output node)}} \tag{1}$$

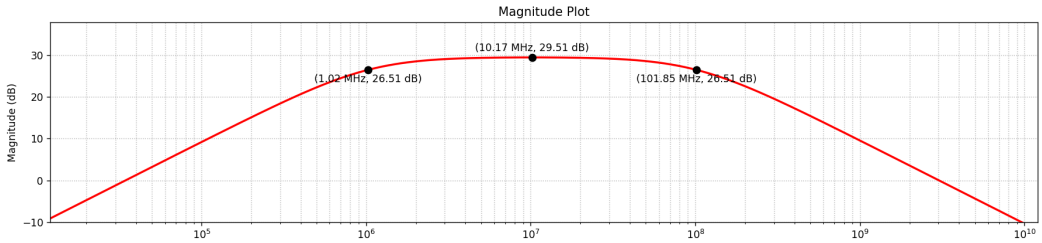


Figure 1: Gain Magnitude Plot from Hand-Calculated Values

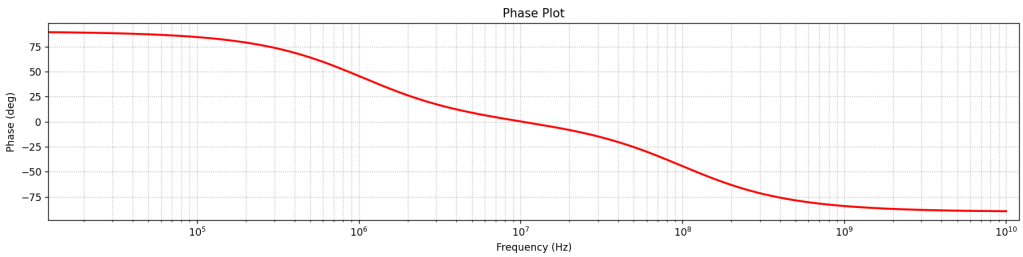


Figure 2: Gain Phase Plot from Hand-Calculated Values

```
PS D:\IITB2\Sem 7\EE618\Assignment 2> python Bode_Plot.py
Max gain = 29.51 dB at 10.2 MHz
f_L (-3 dB) = 1.023 MHz, f_H (-3 dB) = 101.8 MHz, BW = 100.8 MHz
```

Figure 3: Gain and Bandwith Results for Hand Calculations

2 Design-1

2.1 Schematic and Testbench

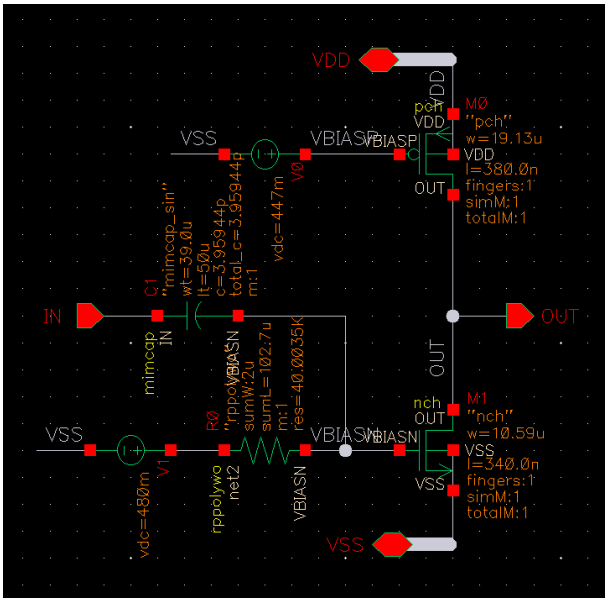


Figure 4: Amplifier Design showing component parameters

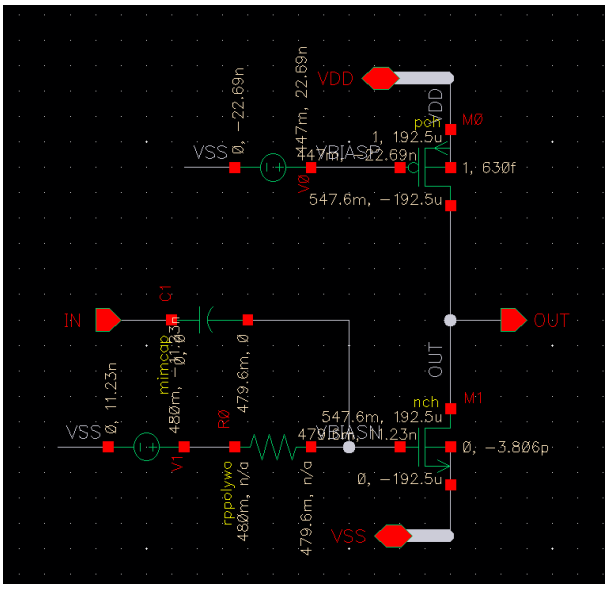


Figure 5: Amplifier Design showing DC Currents and Voltages

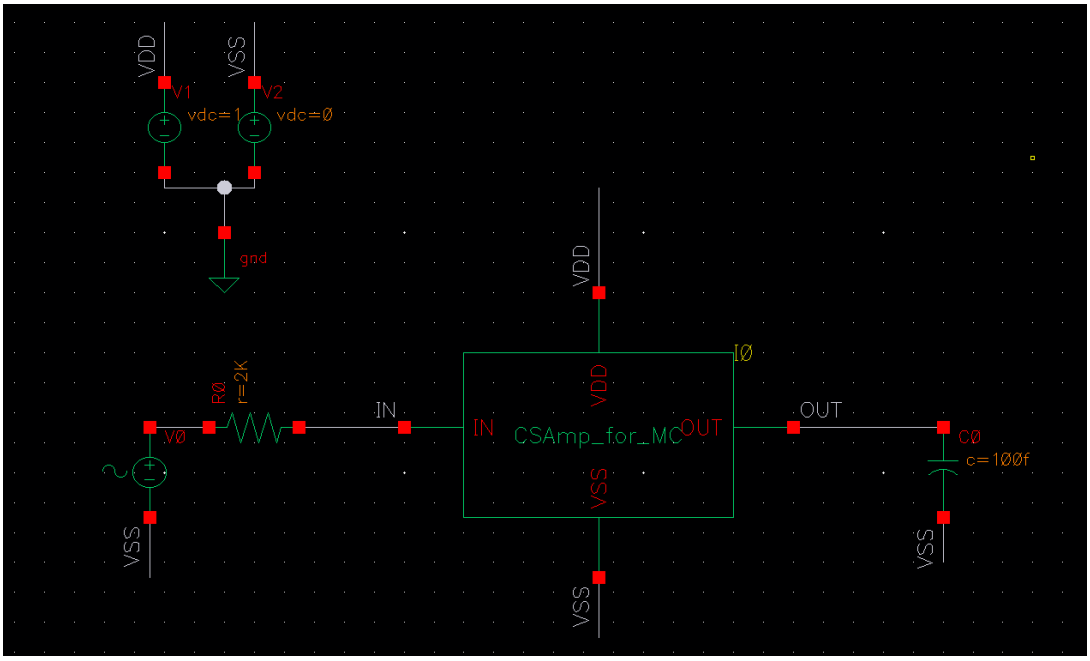


Figure 6: Testbench showing component parameters

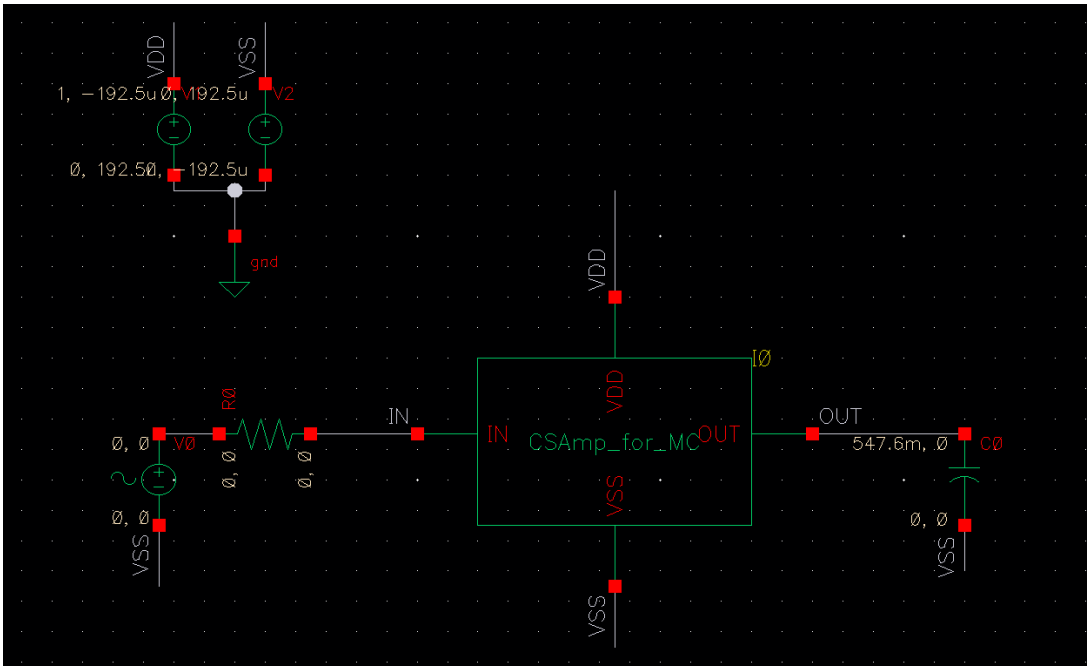


Figure 7: Testbench showing DC Currents and Voltages

2.2 Results of DC Simulation

Outputs Setup | Run Preview

Name	Type	Details	Value	Plot	Plot Target	Save	Spec
I_Bias	expr	IS("/I0/M1/D")	192.5u	☑		☐	
dB20	expr	dB20(VF("/OUT"))		☑		☐	
dB20_at_1M	expr	value(dB20(VF("/OUT")) 1000000)	23.76	☑		☐	
dB20_at_10M	expr	value(dB20(VF("/OUT")) 10000000)	26.45	☑		☐	
dB20_at_100M	expr	value(dB20(VF("/OUT")) 100000000)	23.84	☑		☐	
dB20_Max	expr	ymax(dB20(VF("/OUT")))	26.45	☑		☐	
BW	expr	bandwidth(VF("/OUT") 3 "band")	108.8M	☑		☐	
Phase	expr	phase(VF("/OUT"))		☑		☐	
DC_Vout	expr	VDC("/OUT")	547.6m	☑		☐	
gm	expr	OP("/I0/M1" "gm")	1.959m	☑		☐	
gds_PMOS	expr	OP("/I0/M0" "gds")	43.2u	☑		☐	
gds_NMOS	expr	OP("/I0/M1" "gds")	42.95u	☑		☐	

Figure 8: Results of the DC Simulation

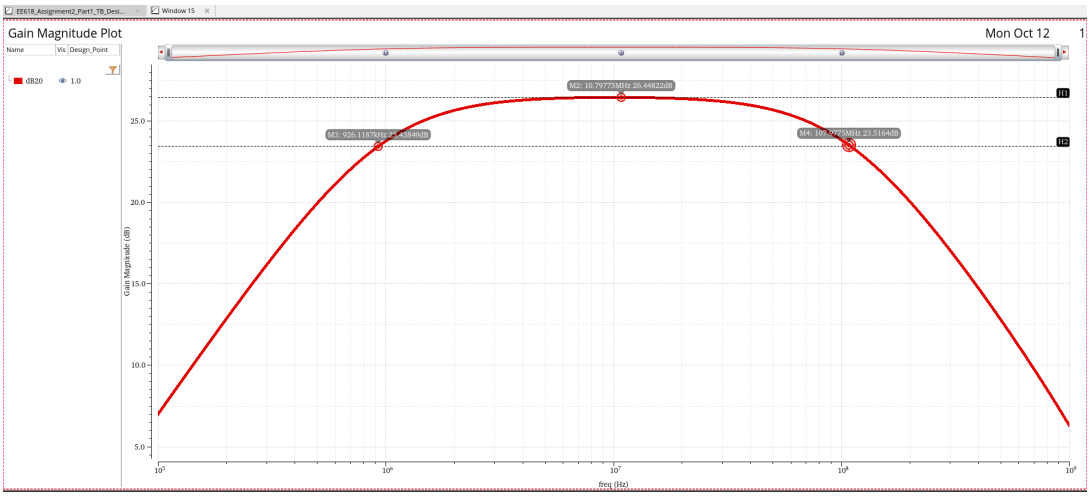


Figure 9: Gain Magnitude Plot (dB vs frequency) with f_L, $f_{maxgain}$ and f_H annotated

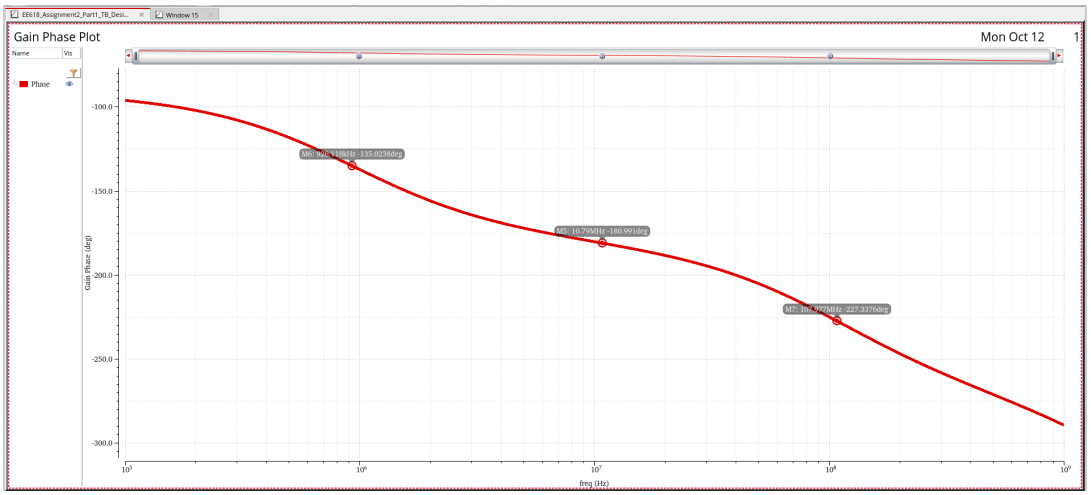


Figure 10: Gain Phase Plot (deg vs frequency) with f_L, $f_{maxgain}$ and f_H annotated

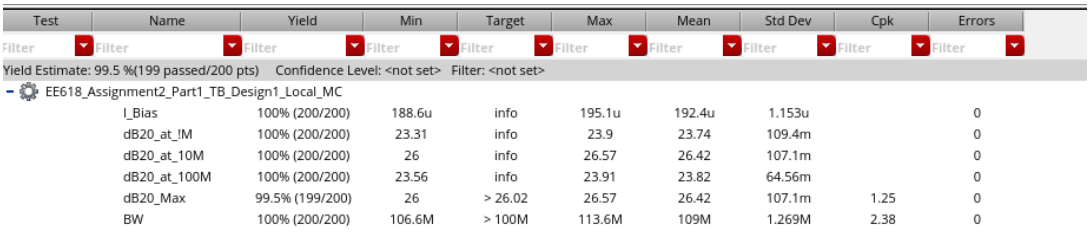
2.3 Requirements

Thus, for Design-1, the following required specifications were met:

- Maximum Small-Signal Gain = 26.45dB

- -3dB Bandwidth = 108.8MHz
- DC Output Voltage = 547.6mV
- f_L = 926.11kHz
- Total DC Power = $V_{DD} * I_D$ = 192.5 μ W

2.4 Results of Local MC Simulation



Test	Name	Yield	Min	Target	Max	Mean	Std Dev	Cpk	Errors
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
Yield Estimate: 99.5 %(199 passed/200 pts)	Confidence Level: <not set>	Filter: <not set>							
EE618_Assignment2_Part1_TB_Design1_Local_MC									
	I_Bias	100% (200/200)	188.6u	info	195.1u	192.4u	1.153u		0
	dB20_at_lM	100% (200/200)	23.31	info	23.9	23.74	109.4m		0
	dB20_at_10M	100% (200/200)	26	info	26.57	26.42	107.1m		0
	dB20_at_100M	100% (200/200)	23.56	info	23.91	23.82	64.56m		0
	dB20_Max	99.5% (199/200)	26	> 26.02	26.57	26.42	107.1m	1.25	0
	BW	100% (200/200)	106.6M	> 100M	113.6M	109M	1.269M	2.38	0

Figure 11: Results for Local MC Simulation

Property	Mean	Std. Dev.
Max. Small-Signal Gain (dB)	26.42	0.107
-3dB Bandwidth (MHz)	109	1.269

Table 1: Statistical analysis (Local MC) Results

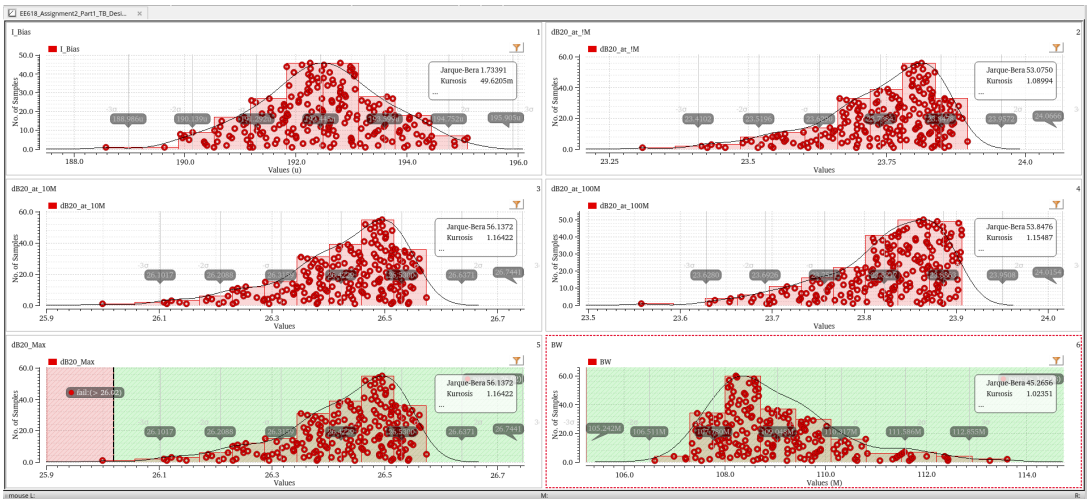


Figure 12: Plots for Local MC Simulation

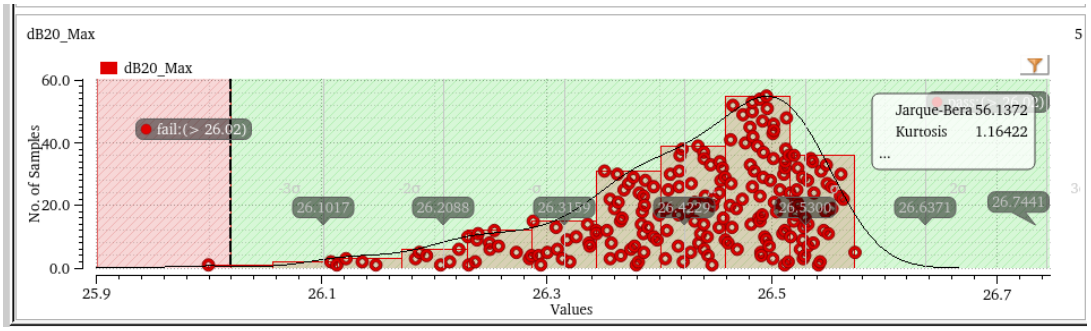


Figure 13: Local MC Simulation for maximum small-signal gain

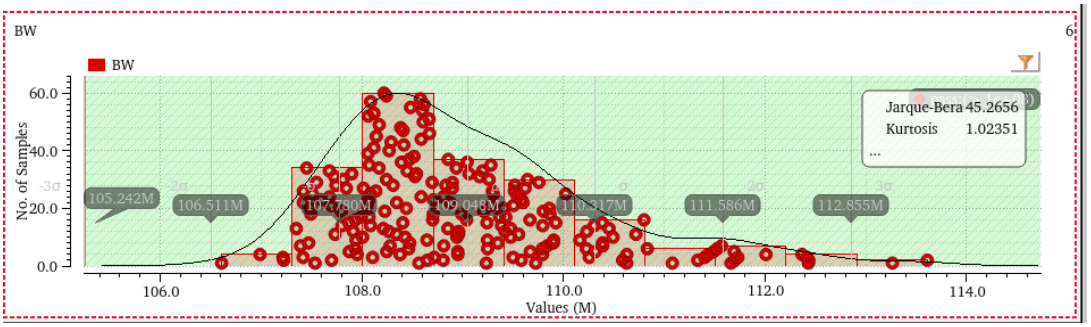
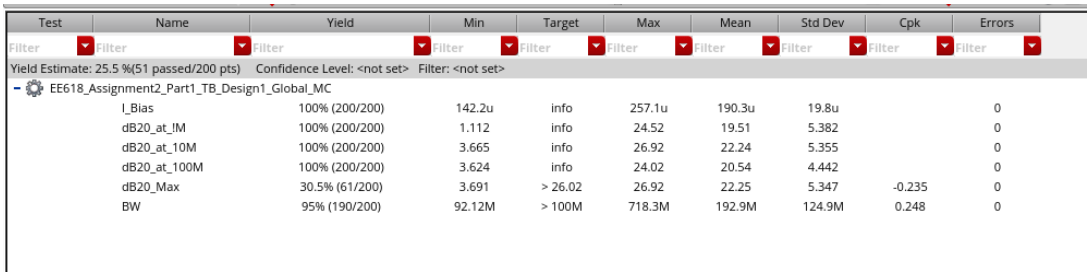


Figure 14: Local MC Simulation for Bandwidth

2.5 Results of Global MC Simulation



Test	Name	Yield	Min	Target	Max	Mean	Std Dev	Cpk	Errors
Yield Estimate: 25.5 %(51 passed/200 pts) Confidence Level: <not set> Filter: <not set>									
EE618_Assignment2_Part1_TB_Design1_Global_MC									
	I_Bias	100% (200/200)	142.2u	info	257.1u	190.3u	19.8u		0
	dB20_at_!M	100% (200/200)	1.112	info	24.52	19.51	5.382		0
	dB20_at_10M	100% (200/200)	3.665	info	26.92	22.24	5.355		0
	dB20_at_100M	100% (200/200)	3.624	info	24.02	20.54	4.442		0
	dB20_Max	30.5% (61/200)	3.691	> 26.02	26.92	22.25	5.347	-0.235	0
	BW	95% (190/200)	92.12M	> 100M	718.3M	192.9M	124.9M	0.248	0

Figure 15: Results for Global MC Simulation

Property	**Mean**	**Std. Dev.**
Max. Small-Signal Gain (dB)	22.25	5.347
-3dB Bandwidth (MHz)	192.9	124.9

Table 2: Statistical analysis (Global MC) Results

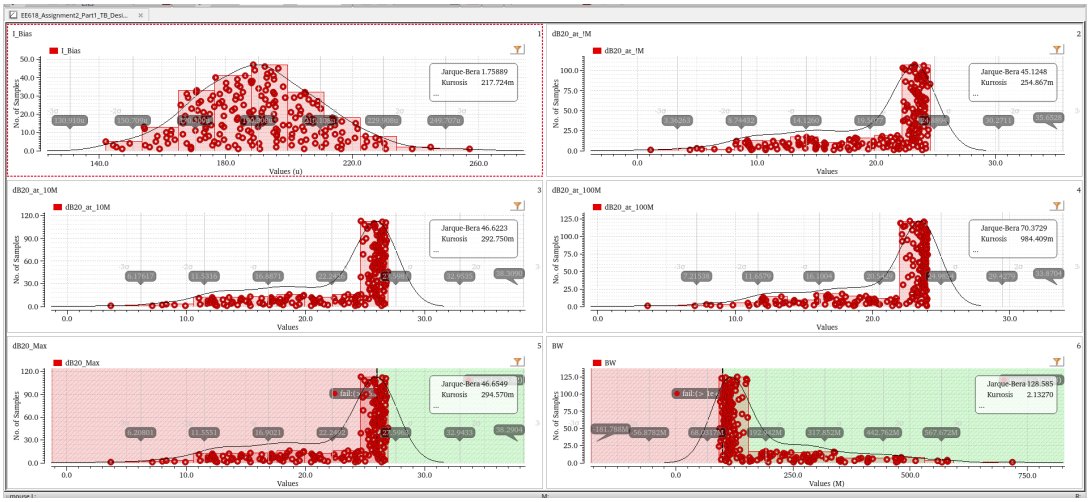


Figure 16: Plots for Global MC Simulation

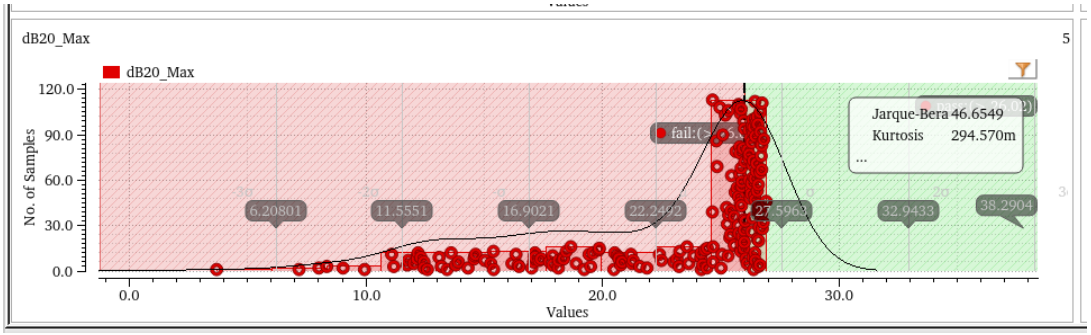


Figure 17: Global MC Simulation for maximum small-signal gain

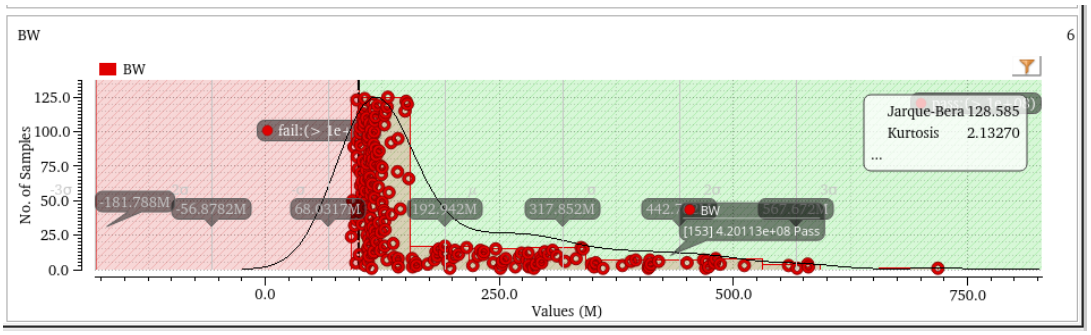


Figure 18: Global MC Simulation for Bandwidth

3 Design-2

3.1 Schematic and Testbench

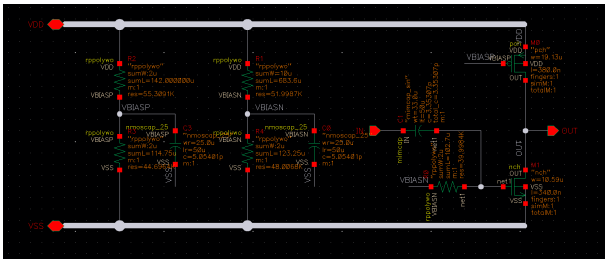


Figure 19: Amplifier Design showing component parameters

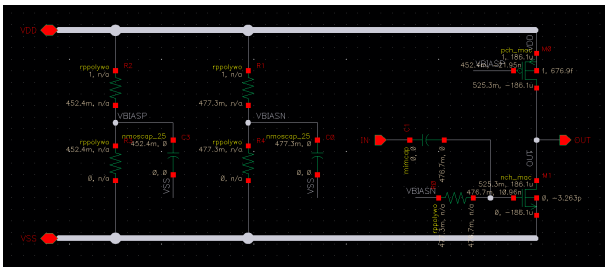


Figure 20: Amplifier Design showing DC Currents and Voltages

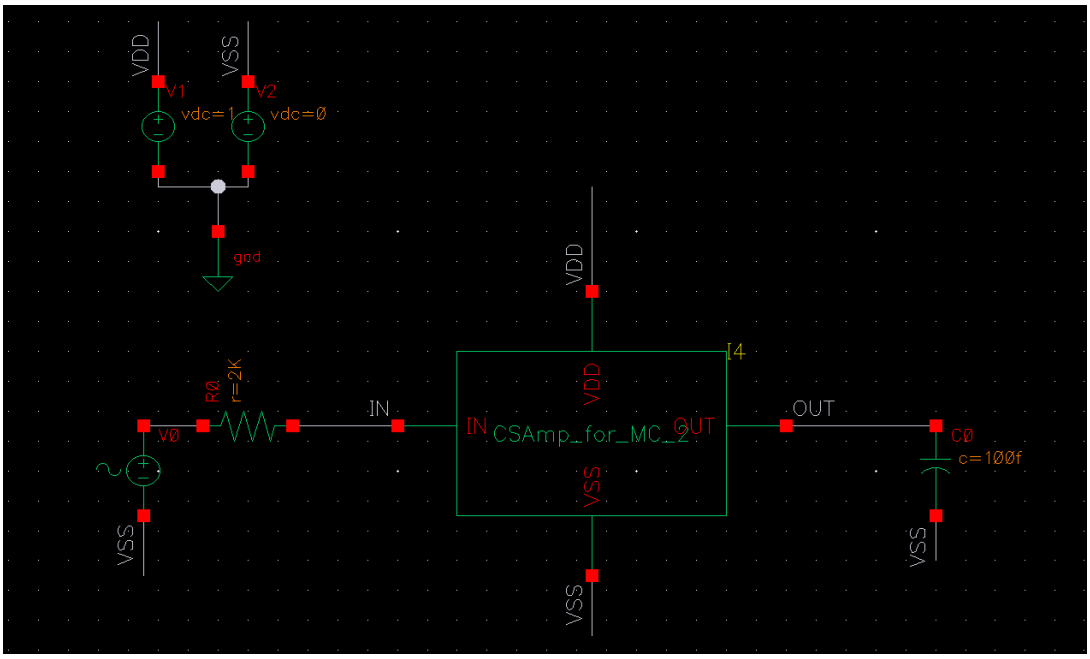


Figure 21: Testbench showing component parameters

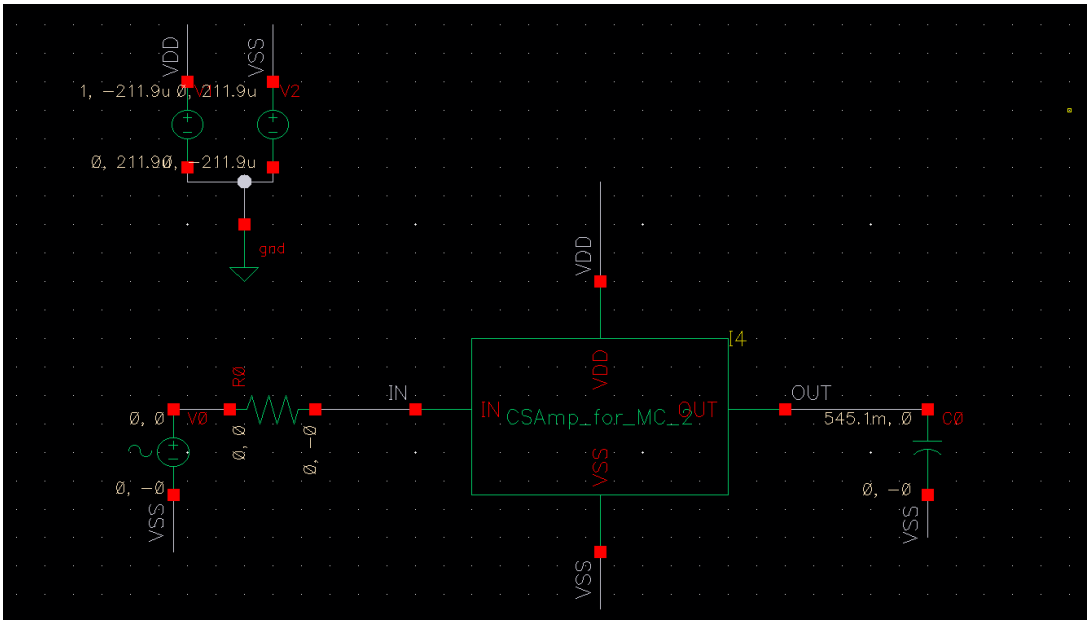


Figure 22: Testbench showing DC Currents and Voltages

3.2 Results of DC Simulation

Name	Type	Details	Value	Plot	Plot Target	Save	Spec
dB20	expr	dB20(VF("/OUT"))		☑		☐	
dB20_at_!M	expr	value(dB20(VF("/OUT")) 1000000)	23.41	☑		☐	
dB20_at_10M	expr	value(dB20(VF("/OUT")) 10000000)	26.28	☑		☐	
dB20_at_100M	expr	value(dB20(VF("/OUT")) 100000000)	23.77	☑		☐	
dB20_Max	expr	ymax(dB20(VF("/OUT")))	26.3	☑		☐	
BW	expr	bandwidth(VF("/OUT") 3 "band")	110.8M	☑		☐	
Phase	expr	phase(VF("/OUT"))		☑		☐	
I_Bias	expr	IS("/I4/M1/D")	191.9u	☑		☐	
DC_Vout	expr	VDC("/OUT")	545.1m	☑		☐	
gm	expr	OP("/I4/M1" "gm")	1.955m	☑		☐	
gds_PMOS	expr	OP("/I4/M0" "gds")	42.84u	☑		☐	
gds_NMOS	expr	OP("/I4/M1" "gds")	42.96u	☑		☐	

Figure 23: Results of the DC Simulation

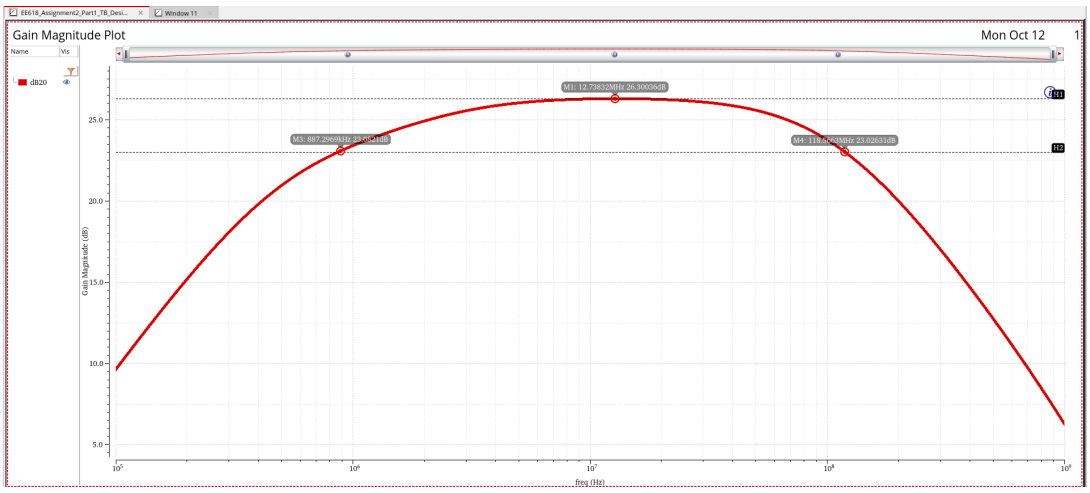


Figure 24: Gain Magnitude Plot (dB vs frequency) with f_L, $f_{maxgain}$ and f_H annotated

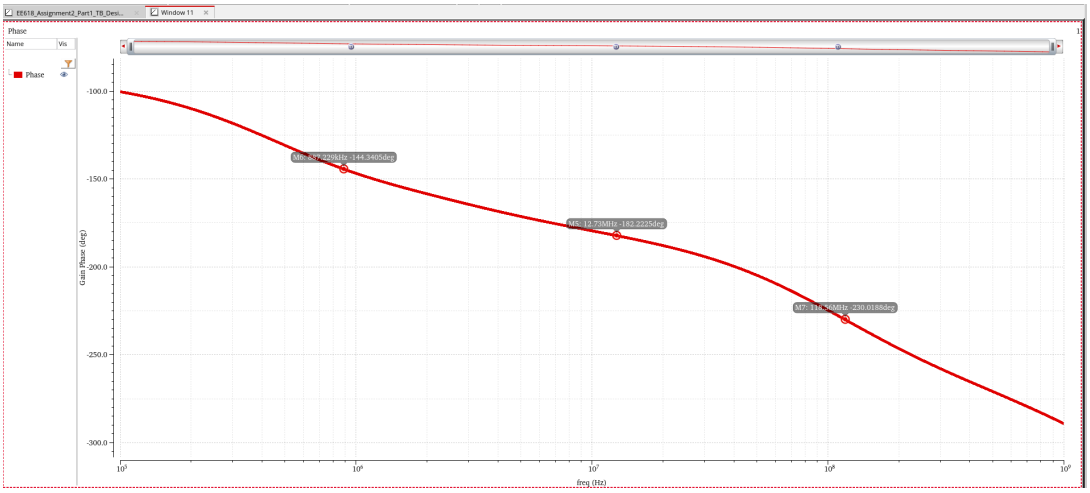


Figure 25: Gain Phase Plot (deg vs frequency) with f_L, $f_{maxgain}$ and f_H annotated

3.3 Requirements

Thus, for Design-2, the following required specifications were met:

- Maximum Small-Signal Gain = 26.3dB
- -3dB Bandwidth = 110.8MHz
- DC Output Voltage = 545.1mV
- f_L = 887.229kHz
- Total DC Power = $V_{DD} * I_D$ = 191.8 μ W + 10 μ W + 10 μ W = 211.8 μ W

3.4 Results of Local MC Simulation

Test	Name	Yield	Min	Target	Max	Mean	Std Dev	Cpk	Errors
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
Yield Estimate: 97 %(194 passed/200 pts)	Confidence Level: <not set>	Filter: <not set>							
EE618_Assignment2_Part1_TB_Design2_Local_MC									
	dB20_at_!M	100% (200/200)	23.01	info	23.53	23.39	106.1m		0
	dB20_at_!0M	100% (200/200)	25.89	info	26.4	26.26	102m		0
	dB20_at_100M	100% (200/200)	23.53	info	23.84	23.75	62.07m		0
	dB20_Max	97% (194/200)	25.91	> 26.02	26.41	26.27	101.4m	0.835	0
	BW	100% (200/200)	108.9M	> 100M	115.3M	111.1M	1.195M	3.1	0
	I_Bias	100% (200/200)	189.1u	info	194.3u	191.9u	1.129u		0

Figure 26: Results for Local MC Simulation

Property	Mean	Std. Dev.
Max. Small-Signal Gain (dB)	26.27	0.1014
-3dB Bandwidth (MHz)	111.1	1.195

Table 3: Statistical analysis (Local MC) Results

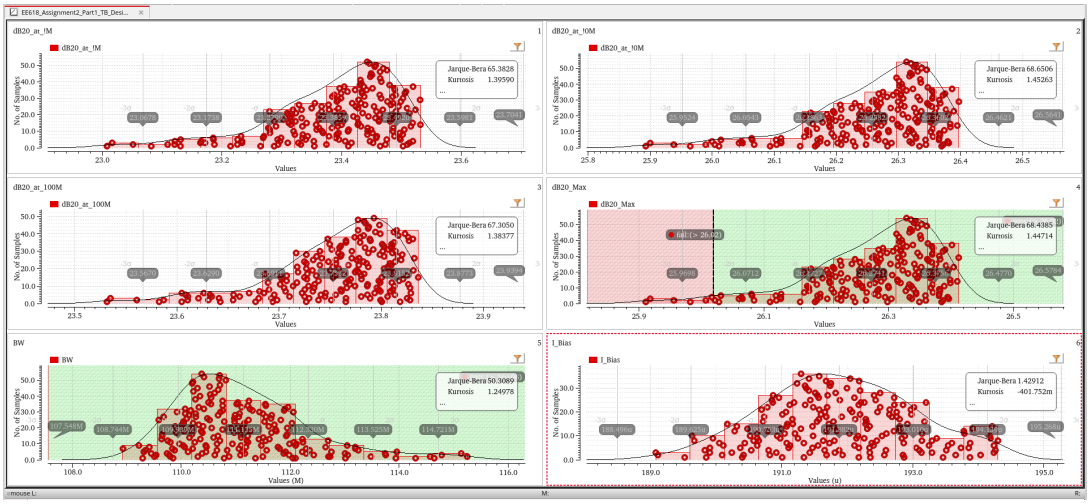


Figure 27: Plots for Local MC Simulation

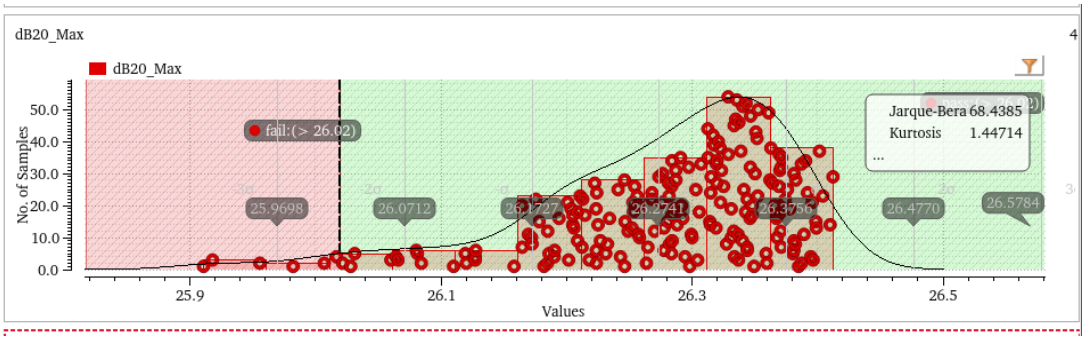


Figure 28: Local MC Simulation for maximum small-signal gain

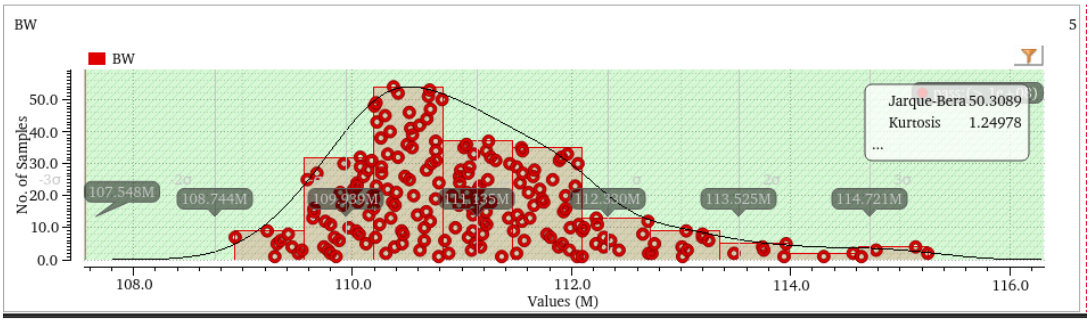


Figure 29: Local MC Simulation for Bandwidth

3.5 Results of Global MC Simulation

Test	Name	Yield	Min	Target	Max	Mean	Std Dev	Cpk	Errors
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
Yield Estimate: 26 %(52 passed/200 pts)	Confidence Level: <not set>	Filter: <not set>							
EE618_Assignment2_Part1_TB_Design2_Global_MC									
	dB20_at_fM	100% (200/200)	118.3m	info	24.34	19.1	5.5		0
	dB20_at_f0M	100% (200/200)	2.992	info	26.76	22.02	5.457		0
	dB20_at_100M	100% (200/200)	3.013	info	23.94	20.41	4.554		0
	dB20_Max	29.5% (59/200)	3.07	> 26.02	26.77	22.06	5.434	-0.243	0
	BW	96.5% (193/200)	92.96M	> 100M	740M	197.1M	128.7M	0.252	0
	I_Bias	100% (200/200)	140.3u	info	254.9u	189.6u	19.73u		0

Figure 30: Results for Global MC Simulation

Property	Mean	Std. Dev.
Max. Small-Signal Gain (dB)	22.06	5.434
-3dB Bandwidth (MHz)	197.1	128.7

Table 4: Statistical analysis (Global MC) Results

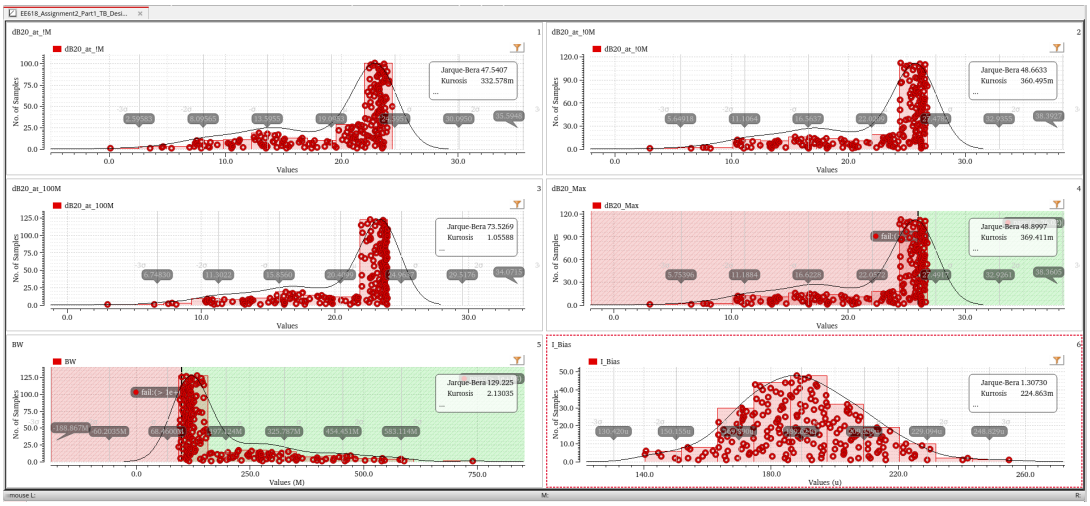


Figure 31: Plots for Global MC Simulation

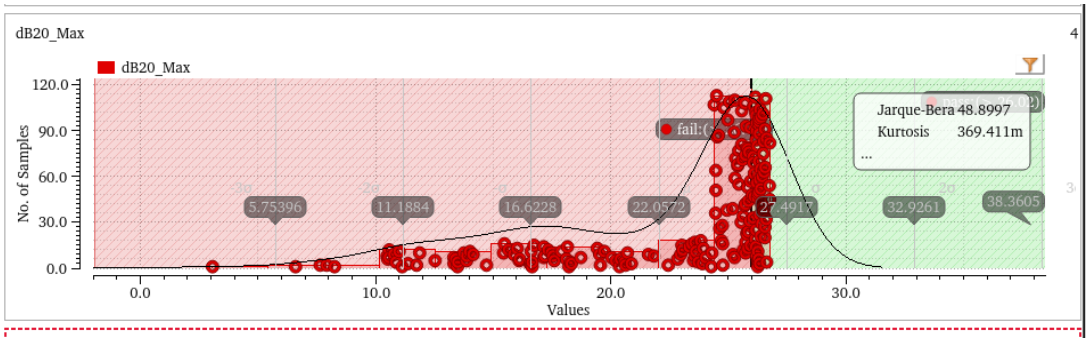


Figure 32: Global MC Simulation for maximum small-signal gain

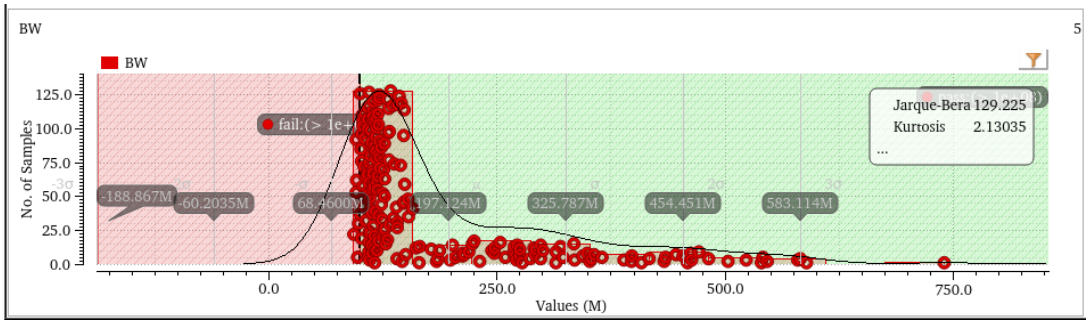


Figure 33: Global MC Simulation for Bandwidth

4 Design-3

4.1 Schematic and Testbench

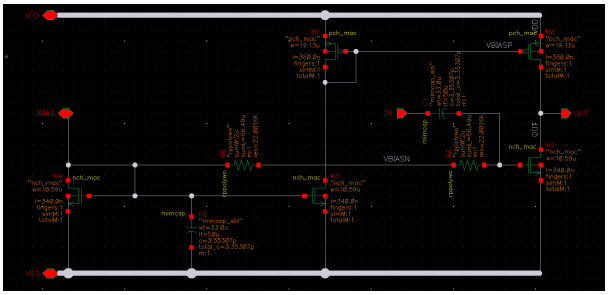


Figure 34: Amplifier Design showing component parameters

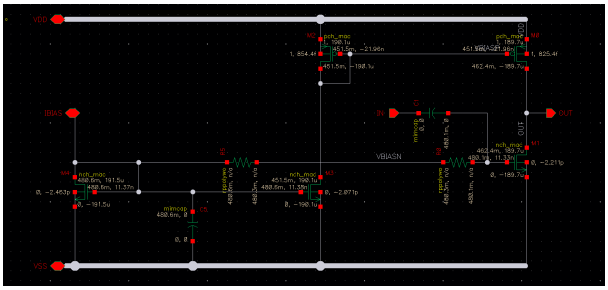


Figure 35: Amplifier Design showing DC Currents and Voltages

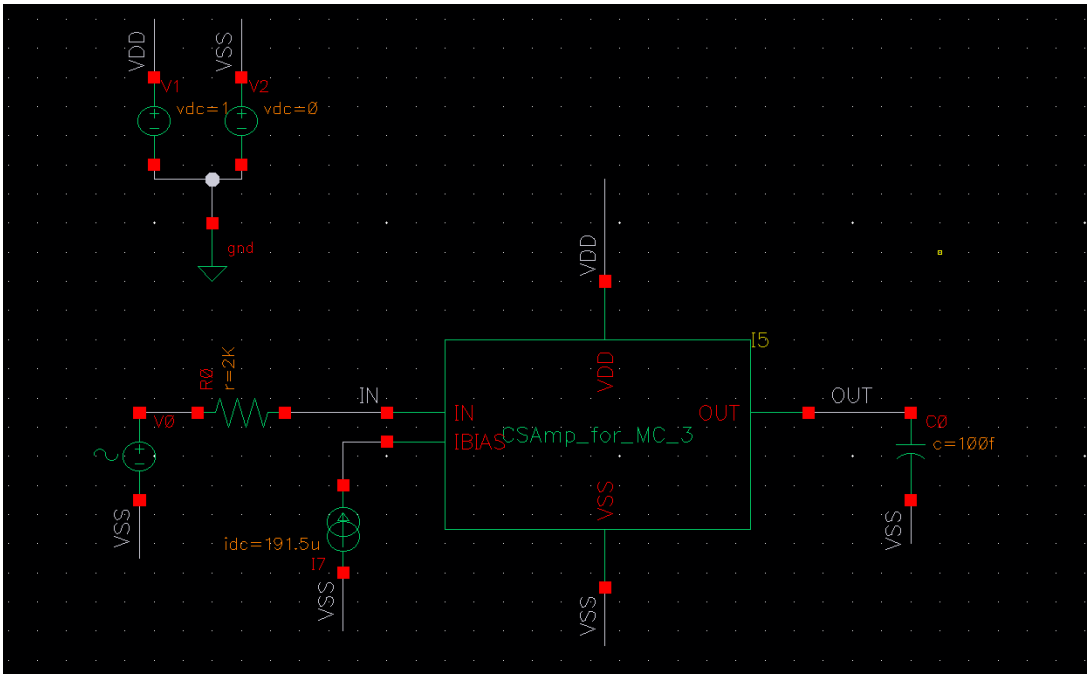


Figure 36: Testbench showing component parameters

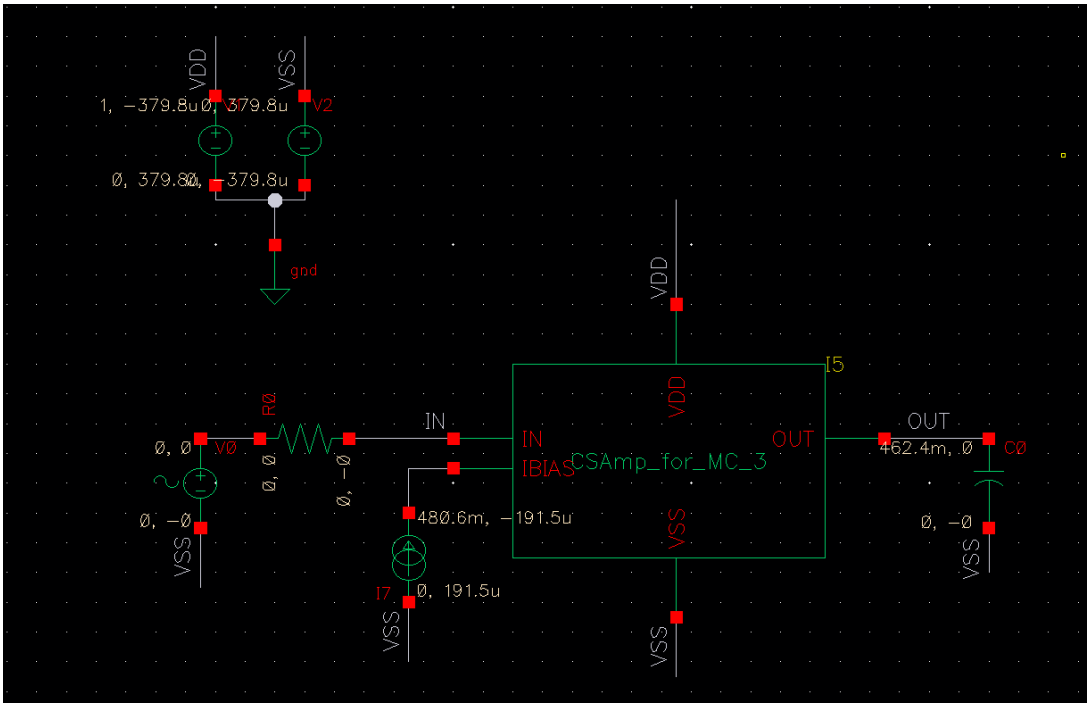


Figure 37: Testbench showing DC Currents and Voltages

4.2 Results of DC Simulation

Outputs Setup | Run Preview

Name	Type	Details	Value	Plot	Plot Target	Save	Spec
dB20	expr	dB20(VF("/OUT"))					
dB20_at_1M	expr	value(dB20(VF("/OUT")) 1000000)	23.58				
dB20_at_10M	expr	value(dB20(VF("/OUT")) 10000000)	26.51				
dB20_at_100M	expr	value(dB20(VF("/OUT")) 100000000)	23.72				
dB20_Max	expr	ymax(dB20(VF("/OUT")))	26.51				
BW	expr	bandwidth(VF("/OUT") 3 "band")	104.1M				
Phase	expr	phase(VF("/OUT"))					
I_Bias	expr	IS("/I5/M1/D")	189.7u				
DC_Vout	expr	VDC("/OUT")	462.4m				
gm	expr	OP("/I5/M1" "gm")	1.932m				
gds_PMOS	expr	OP("/I5/M0" "gds")	36.49u				
gds_NMOS	expr	OP("/I5/M1" "gds")	46.92u				

Figure 38: Results of the DC Simulation

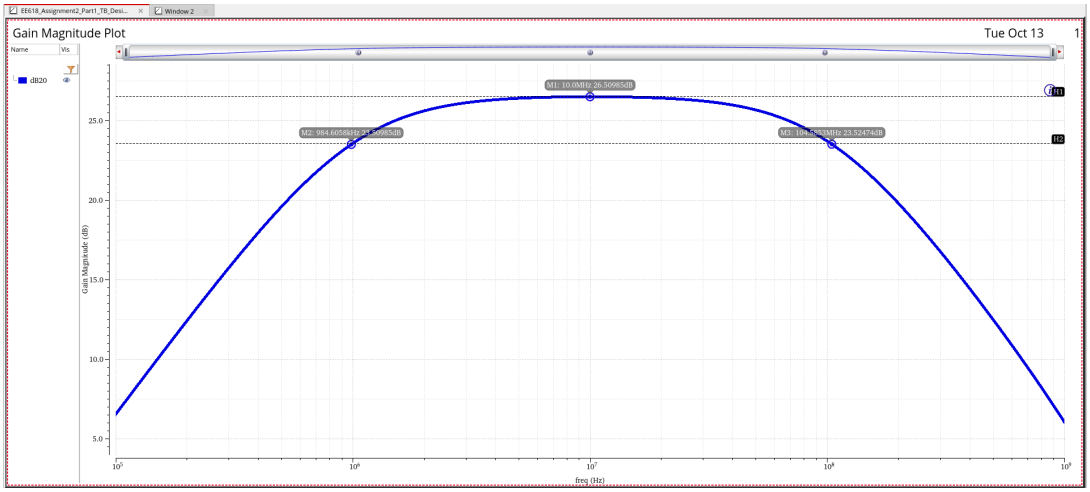


Figure 39: Gain Magnitude Plot (dB vs frequency) with f_L, $f_{maxgain}$ and f_H annotated

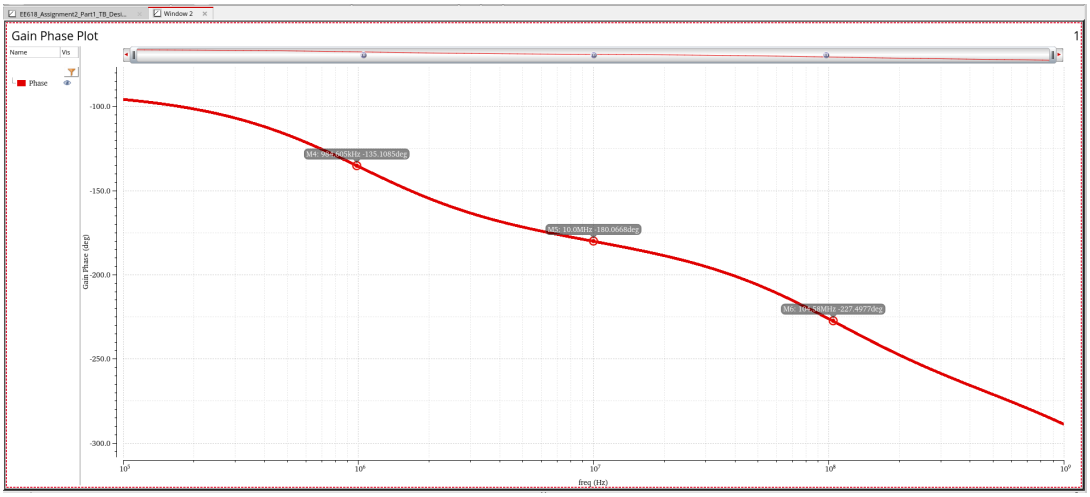


Figure 40: Gain Phase Plot (deg vs frequency) with f_L, $f_{maxgain}$ and f_H annotated

4.3 Requirements

Thus, for Design-3, the following required specifications were met:

- Maximum Small-Signal Gain = 26.51dB
- -3dB Bandwidth = 104.1MHz
- DC Output Voltage = 462.4mV
- f_L = 984.6kHz
- Total DC Power = $V_{DD} * I_D$ = 189.7 μW + 190.1 μW = 379.8 μW (If power from I_{BIAS} is also to be included, then 571.9 μW)

4.4 Results of Local MC Simulation

Test	Name	Yield	Min	Target	Max	Mean	Std Dev	Cpk	Errors
		Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
Yield Estimate: 100 %(200 passed/200 pts)	Confidence Level: <not set>	Filter: <not set>							
EE618_Assignment2_Part1_TB_Design3_Local_MC									
	dB20_at_!M	100% (200/200)	23.09	info	23.67	23.54	73.93m		0
	dB20_at_!0M	100% (200/200)	26.03	info	26.59	26.47	71.8m		0
	dB20_at_100M	100% (200/200)	23.4	info	23.77	23.7	46.27m		0
	dB20_Max	100% (200/200)	26.03	> 26.02	26.59	26.47	71.8m	2.1	0
	BW	100% (200/200)	101M	> 100M	108.4M	104.5M	1.3M	1.14	0
	I_Bias	100% (200/200)	181.8u	info	197.8u	189.6u	2.545u		0

Figure 41: Results for Local MC Simulation

Property	Mean	Std. Dev.
Max. Small-Signal Gain (dB)	26.47	0.0718
-3dB Bandwidth (MHz)	104.5	1.3

Table 5: Statistical analysis (Local MC) Results

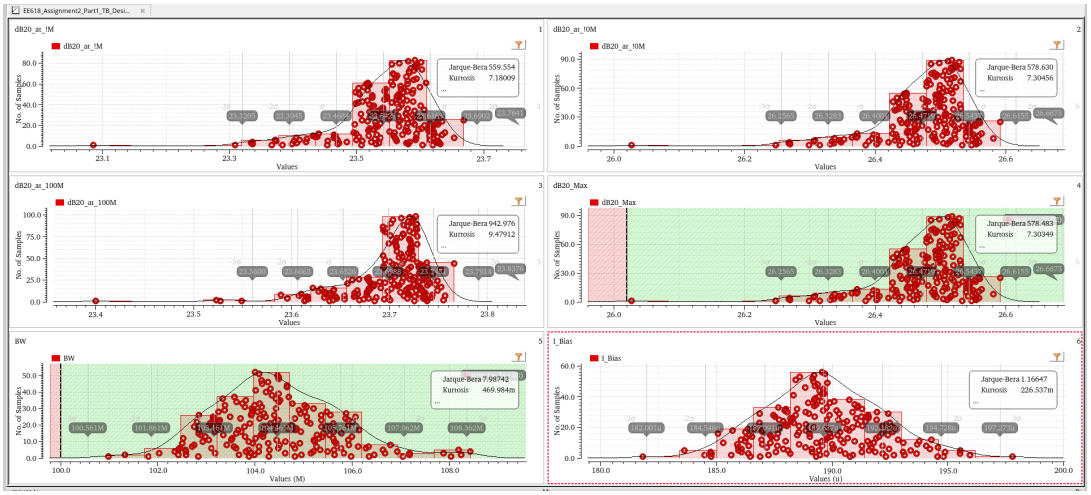


Figure 42: Plots for Local MC Simulation

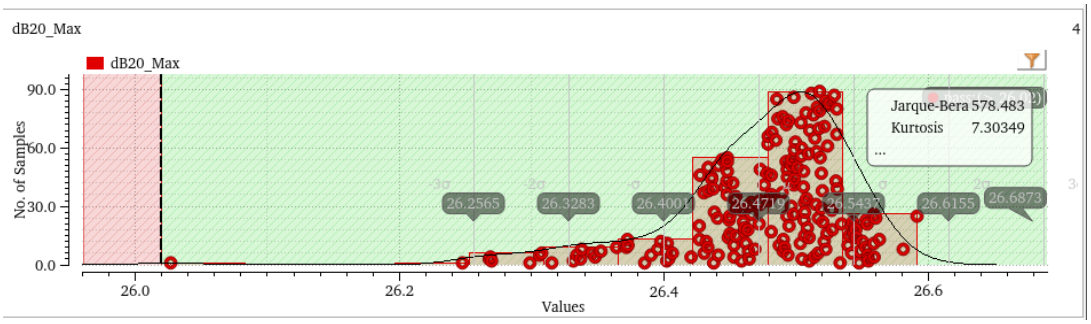


Figure 43: Local MC Simulation for maximum small-signal gain

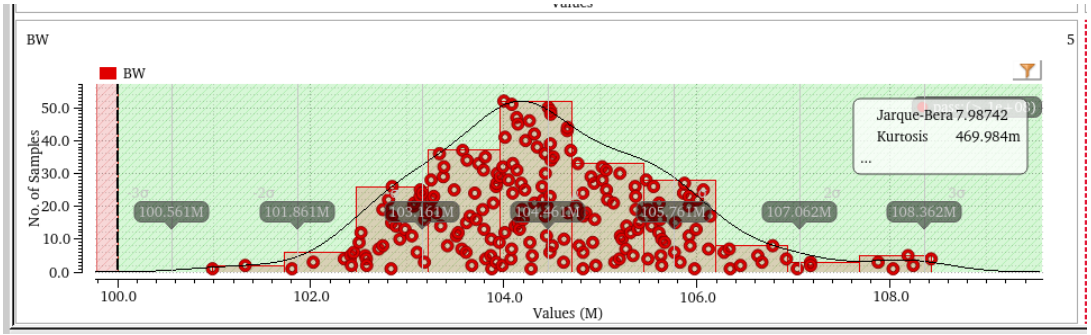
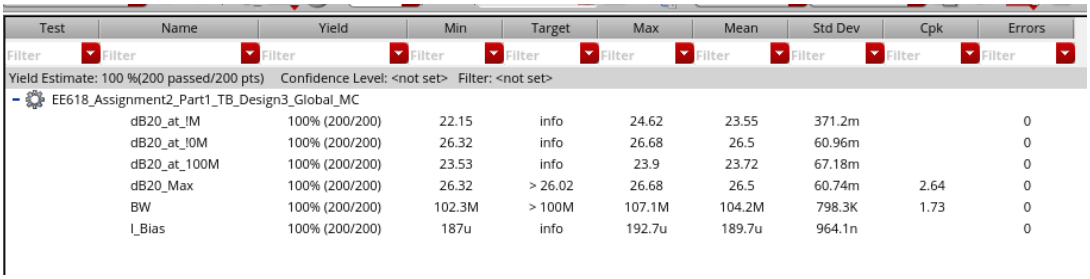


Figure 44: Local MC Simulation for Bandwidth

4.5 Results of Global MC Simulation



Test	Name	Yield	Min	Target	Max	Mean	Std Dev	Cpk	Errors
Yield Estimate: 100 %(200 passed/200 pts) Confidence Level: <not set> Filter: <not set>									
EE618_Assignment2_Part1_TB_Design3_Global_MC									
	dB20_at_!M	100% (200/200)	22.15	info	24.62	23.55	371.2m		0
	dB20_at_!0M	100% (200/200)	26.32	info	26.68	26.5	60.96m		0
	dB20_at_100M	100% (200/200)	23.53	info	23.9	23.72	67.18m		0
	dB20_Max	100% (200/200)	26.32	> 26.02	26.68	26.5	60.74m	2.64	0
	BW	100% (200/200)	102.3M	> 100M	107.1M	104.2M	798.3K	1.73	0
	I_Bias	100% (200/200)	187u	info	192.7u	189.7u	964.1n		0

Figure 45: Results for Global MC Simulation

Property	Mean	Std. Dev.
Max. Small-Signal Gain (dB)	26.5	0.06074
-3dB Bandwidth (MHz)	104.2	0.7983

Table 6: Statistical analysis (Global MC) Results

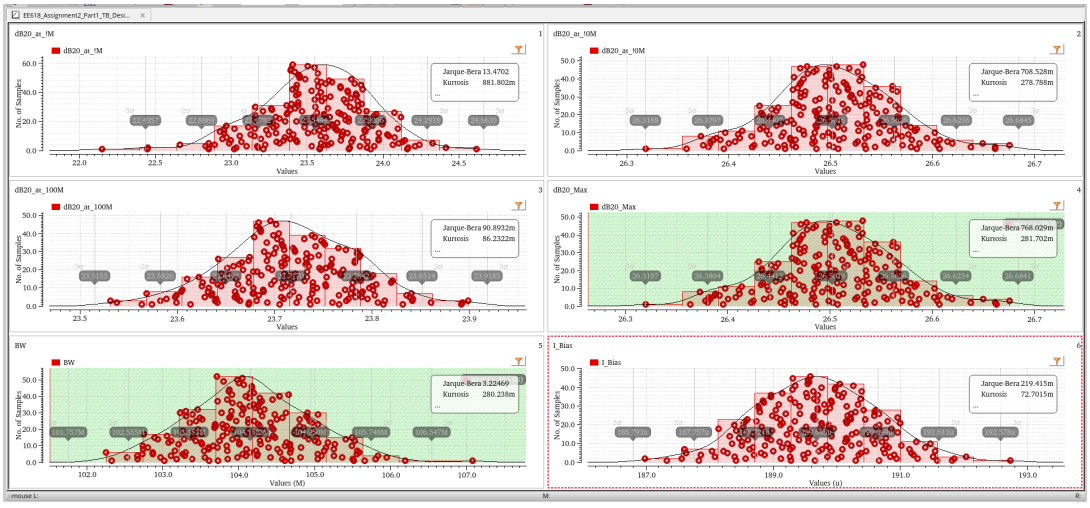


Figure 46: Plots for Global MC Simulation

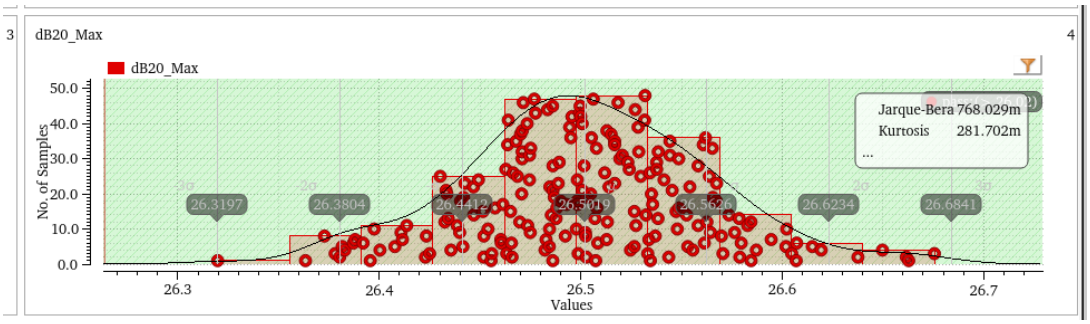


Figure 47: Global MC Simulation for maximum small-signal gain

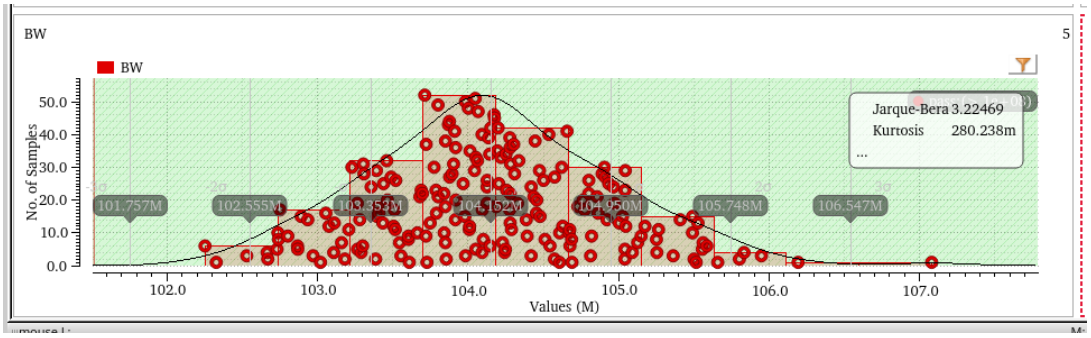


Figure 48: Global MC Simulation for Bandwidth

5 Some Final Points

5.1 Monte Carlo Results

- **Local MC:** All three designs are close to nominal ($\sigma_{gain} < 0.11$ dB, $\sigma_{BW} \approx 1.2$–1.3 MHz). Gain yield is 99.5%, 97% and 100% for Designs 1, 2 and 3, and BW yield is 100% for all three.

- **Global MC, Designs 1 and 2:** Gain yield is only about 30% (mean ≈ 22 dB, $\sigma \approx 5.4$ dB). Their gate voltages are fixed, so a process shift in V_{th} changes I_D directly ($\sigma(I_D) \approx 20$ μA). The NMOS and PMOS currents then no longer balance, and the output node, which has a high impedance, drifts until a transistor leaves saturation.

- **Global MC, Design 3:** It is the most robust, with 100% yield, gain 26.50 ± 0.061 dB and BW 104.2 ± 0.80 MHz. The current mirror forces I_D to follow the ideal reference ($\sigma(I_D) \approx 1$ μA), so V_{ov}, g_m and g_{out} track the process, and its nominal gain margin is also the largest.

- **Trade-off:** Design 3 needs extra bias branches and so uses more power than Design 1. Design 2's resistive divider adds power without added robustness, because its bias does not track the transistor parameters.

5.2 Power Minimisation

- **Principle:** $P_{DC} = V_{DD} \cdot I_{VDD}$ is minimised by using the lowest I_D that still meets the gain (> 26.02 dB) and bandwidth (> 100 MHz) specs. Since $f_H \propto I_D/C_{L,tot}$, the bandwidth sets this lower limit on the current, which is why the hand calculation gives $I_D \approx 194$ μA.

- **Design 1 and 2:** Design 1 uses only the single amplifier branch (≈ 192.5 μW). Design 2 adds resistive divider branches, and these cost extra power without improving robustness.

- **Design 3 (mirror ratios):** To reduce the power, I tried different mirror ratios (2:1, 4:1, 8:1) to scale down the reference and the bias branches. Reducing one branch current either raised the currents in the other branches or left the amplifier branch without enough current, so the gain or bandwidth specification failed. A 1:1 ratio gave the best compromise, with about 190 μA in each branch.

- **Final result:** Design 3 draws ≈ 380 μA from V_{DD} ($P \approx 380$ μW), where the ideal reference current is counted only once because it is already part of the supply current. This is the lowest power at which Design 3 meets all specifications in simulation, and the extra power compared with Design 1 is the price of its robustness to process variation.

6 Conclusion

1. Designed a common-source amplifier by selecting appropriate transistor dimensions and bias voltages to achieve the desired gain and bandwidth specifications.

2. Performed hand calculations and Cadence simulations to analyze the amplifier's DC operating point, small-signal gain, and frequency response. For all three designs, the following required specifications were met:

   - Maximum Small-Signal Gain $>$ 26.02dB
   - -3dB Bandwidth $>$ 100MHz
   - DC Output Voltage $\approx$ 500mV
   - $f_L \approx$ 1MHz (always ensure $>$ 885kHz)
   - Minimum Total DC Power

3. Compared theoretical predictions with simulation results and analyzed deviations arising from practical transistor characteristics and parasitic effects.

4. Evaluated the impact of process and mismatch variations on amplifier performance using Monte-Carlo simulations for different biasing configurations.

5. Analyzed the trade-offs between gain, bandwidth, power consumption, and robustness to variations to identify an efficient and reliable design.